CCH Tagetik

General Data Protection Regulation

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Document control

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1. What is the General Data Protection Regulation?

GPDR is a regulation aiming to protect Personal Data, such as a person's name, surname, email address or phone number.
It addresses both the way data is processed, but also how data is stored / persisted.
It also states the individuals whose Personal Data is stored within an informative system, can request their data to be logically forgotten, as if they never interacted with the system in the first place.
A CCH Tagetik system can be GDPR compliant with respect to encrypting Personal Data on the database.
This protects the confidentiality of such data from a malicious user gaining access to the database, be it from physically accessing the database file, or by executing SQL statements against it in any other way.

2. CCH Tagetik Software support

Starting from version 5.3.SP4, the application supports GPDR compliance in all those functionalities where Personal Data is known to be stored in the Database. This excludes, for instance, usages of Personal Data in Custom Dimensions and any data in the Analytical Workspace.
This applies to both encryption in the database and the possibility to exercise the right to be forgotten for a person.

3. Configuration

With respect to database level encryption, you must decide whether you want encryption to be applied or not, depending on your requirement.
Data encryption configuration can be changed at any time but it that case an Application Server restart will be required.
We strongly advise you to enable data encryption for the protection of your users' Personal Data.
See below for Configuration for Encrypted Data or Configuration for Un-Encrypted Data.

3.1 After Configuration

Starting from version 5.3.SP6, the status of GDPR encryption configuration can be retrieved opening the "Information" window accessible from the user menu in the top right corner from the Legacy UI, or from the Settings menu in the top bar (wrench icon) of the latest UI.
Once the system is configured to support encryption, any modification to Personal Data will be persisted to the database in encrypted form.
In order to have existing data encrypted, after an installation, you are strongly encouraged to run the "Encrypt Data" tool.
This is accessible from "Repository | Tools | Encrypt Data" from the Legacy UI, or from the three points menu in the latest UI.
In order to have existing data un-encrypted instead you can use the "Decrypt Data" tool, accessible from "Repository | Tools | Decrypt Data" from the Legacy UI, or from the three points menu in the latest UI.

3.2 Configuration for Encrypted Data

In order to apply database level data encryption, the system must be provided an encryption key.

The encryption key must be stored in a standard Java Keystore.

A Keystore has a general password that is needed to access it. Furthermore, every entry in the Keystore has an alias and an optional password to access it.

You need to provide the system with the location of the Keystore, general password, key alias and key password. All of these configuration settings can either be:

1. stored in a standard Java properties file, whose location you need to provide to the system by means of a System Property; or
2. they can be all configured via System Properties.

Please refer to the next two sections for details on how to specify these configuration settings.

3.2.1 Properties File Configuration

1. Create a KeyStore.
2. Create properties file with below properties
    a. `keystore.path=`<Path of the Keystore File>
    b. `keystore.storepwd=`<Password of the Keystore>
    c. `keystore.alias=`<Allias for the Key>
    d. `keystore.keypwd=`<Password for the Key>
3. Set the System property on the Application Server
    a. `it.grupposervizi.easy.keystore.properties.path=`<Path of the Properties File>

3.2.2 System Properties Configuration

1. Create a KeyStore.
2. Set the Systems Properties on the Application Server
    a. `it.grupposervizi.easy.keystore.path=`<Path of the Keystore File>
    b. `it.grupposervizi.easy.keystore.storepwd=`<Password of the Keystore>
    c. `it.grupposervizi.easy.keystore.alias=`<Alias for the Key>
    d. `it.grupposervizi.easy.keystore.keypwd=`<Password for the Key>

3.2.3 System Properties Configuration Without Keystore (starting from SP29)

1. Create a key with encryption AES 128 and encode it in base64.
2. Set the System Property on the Application Server
    a. `it.grupposervizi.easy.encryption.key.aes128=`<Key encrypted in AES 128 and encoded in Base64>

3.3 Configuration for Un-Encrypted Data

In case you do not want to use encryption, you need explicitly specify this to the system, by setting the System Property on the Application Server

- `it.grupposervizi.easy.encryption.disabled=true`

3.4 Logging

In order to check whether CCH Tagetik is running with encryption enabled or not a log entry for category `com.tagetik.security.encryption.internal.EncryptionKeyCheckerContextListener` will be written to the log files.

In case where no configuration for encryption is provided and the property for explicitly disabling encryption is not specified or set to false, the application will log a warning message.

4. Restrictions

4.1 Ability to transfer databases

- Once you've implemented encryption, you can no longer copy a physical database across different Application Servers, if they don't share the same keystore. This includes, for instance, migrating data from a DEV environment to PROD.
  The keystore could be shared across different Application Servers. In the case where the keystore isn't shared, sharing a Database will not work either via Database backup, or application Import / Export.
- The CCH Tagetik Database Export feature will export encrypted data for columns which contain Personal Data.

4.2 Technical Limitations

- Different keystores cannot be used for the same database.
- Once encryption has been applied on the database using any of the encryption configurations (file property configuration or system property configuration) disabling encryption using the developer configuration would result to all the encrypted fields to be hardcoded with "Encrypted Text" losing all the previous information. After disabling, whenever you modify "Encrypted Text" field it would be saved as plain text.
- On a system that encrypts users' data, we should strongly suggest not to use the ETL feature to write into the REP_UTENTE table. If such an ETL job is run, then the tool to encrypt data should be run again.

5. Right to be forgotten

This feature is available regardless of the encryption configuration and no installation steps are needed in order to enable this feature.

6. Appendix

6.1 Creating a keystore

In order to create a keystore, the Java `keytool` utility can be used. It is installed by default by the JDK under the path `JAVA_HOME/bin/keytool`. On Windows system the tool is named keytool.exe.

A typical usage would be

```
$ keytool -genseckey -keystore <your-keystore.jck> -storetype jceks -storepass <storepwd> -
keyalg AES -keysize 128 -alias <alias> -keypass <keypwd>
```

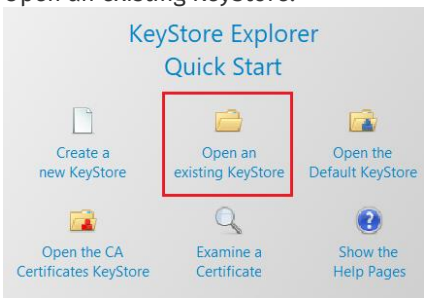
- `-keystore` specifies the path to the keystore. File is created if not present.
- `-storepass` is the password for the whole keystore.
- `-alias` is the alias for the key you're creating.
- `-keypass` is the password for the key you're creating.
- `-keyalg` and `-keysize` specify the type and size of the encryption algorithm for which the key is created. For GDPR related encryption these must be set to `AES` and `128` respectively.

6.2 Creating a key with encryption AES 128 and encoding it in base64

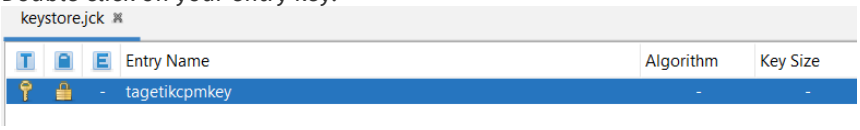
In order to create a key, it's possible to generate an AES 128 encryption key and encode it in base64 using online tools, CLI tools or customized tools (code snippets).

Alternatively, you can follow these steps:

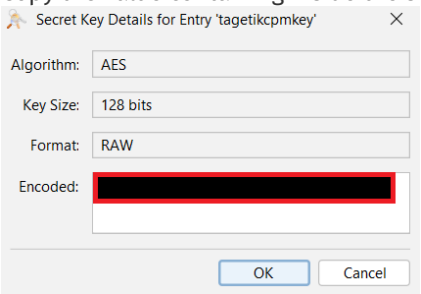
1. Create a KeyStore.
2. Download KeyStore Explorer.
3. Open KeyStore Explorer.
4. Open an existing KeyStore:



5. Choose your keystore.
6. Double click on your entry key:



7. Copy the value containing inside the section "Encoded":



8. Convert the value copied in base64 utilizing online tools or code snippets.

7. Notes and Copyright

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